

```
%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;
scuts_path = 'e:\SHARE\Fe\Data\analysis\07_fit_with_multicut_allJ\sel_cuts';
Cutfitfile = fullfile(scuts_path, 'NG_FFCutsFits.mat');
```

```
if ~exist('NG_FFCutsFits', 'var')
    ld = load(Cutfitfile);
    NG_FFCutsFits = ld.fit_src_struct;
end
```

```
fn = fieldnames(NG_FFCutsFits);
nf = 11
```

```
nf =
11
```

```
%for i = 11:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei800_off200NGmaxFF'
```

```
selectCuts =
'Ei800_off200NGmaxFF'
```

```
if contains(selectCuts, 'msk')
    error('Can not fit field: %s', selectCuts);
end
if contains(selectCuts, 'all_fit_par')
    error('Can not fit field: %s', selectCuts);
end

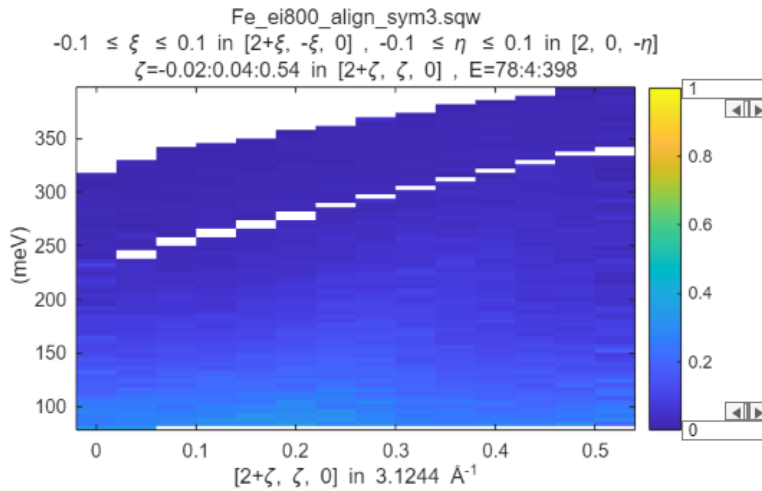
fprintf('Processing cuts: %s \n', selectCuts);
```

```
Processing cuts: Ei800_off200NGmaxFF
```

```
%selectCuts= 'Ei400_off200GPmaxFF';
cuts_list = NG_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts, 'Ei200')
    mi = maps_instrument(200, 600, 'S');
end
if contains(selectCuts, 'Ei400')
    mi = maps_instrument(401, 600, 'S');
end
if contains(selectCuts, 'Ei800')
    mi = maps_instrument(786, 600, 'S');
end
sample=IX_sample(true, [1,0,0], [0,1,0], 'cuboid', [0.04,0.03,0.02]);
for j=1:numel(cuts_list)
```

```
cuts_list{j} = cuts_list{j}.set_instrument(mi);
cuts_list{j} = cuts_list{j}.set_sample(sample);
plot(cuts_list{j}); lz 0 1; keep_figure;
```

end



```
dir_name = "NG";

dE_step = 2; %original energy transfer step data were binned to. No point in going
finer
half_dE = 5; % half width of data binning

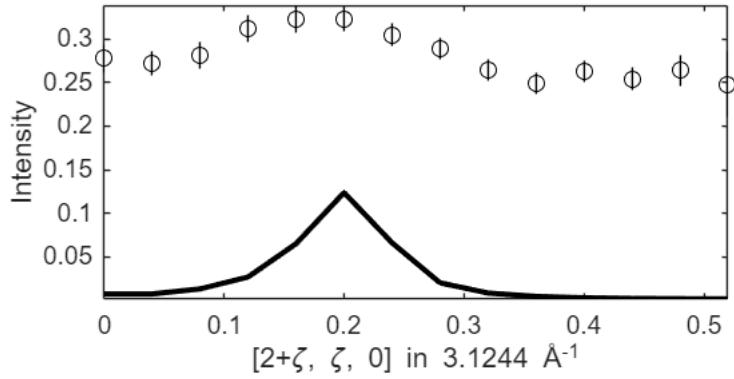
en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(50,en_mm(1));
%en_mm(1) = 100;
en_mm(2) = 380;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
NG_FFCutsFits.(selectCuts) = cuts_list;

NG_FFCutsFits = fit_multicutsFF_along_direction(...
    NG_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

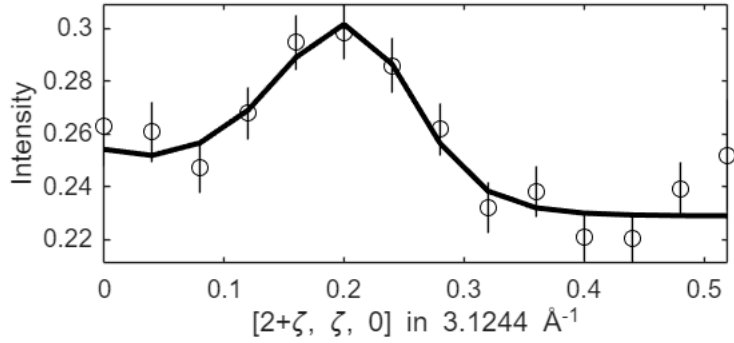
```
*****
**** En = 83±5 going to 373
*****
```

Fe_ei800_align_sym3.sqw
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 78 :
 $\zeta = -0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

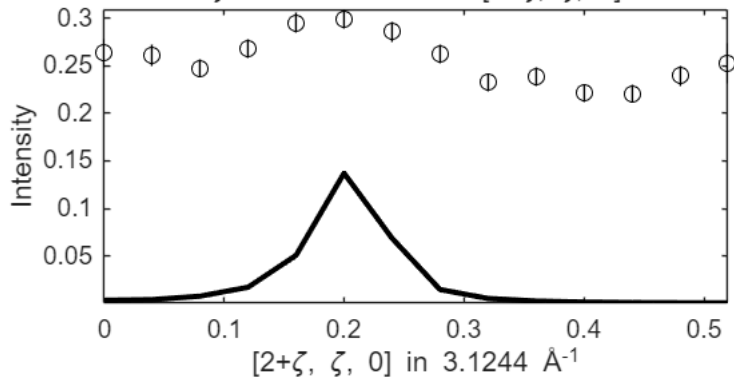


 **** En = 93±5 going to 373

Fe_ei800_align_sym3.sqw
 $:0.65 \pm 0.053$; $J_0 = 29 \pm 1.3$; $J_1 = 0 \pm 0$; $\gamma = 15.8795 \pm 1$
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 88 :
 $\zeta = -0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



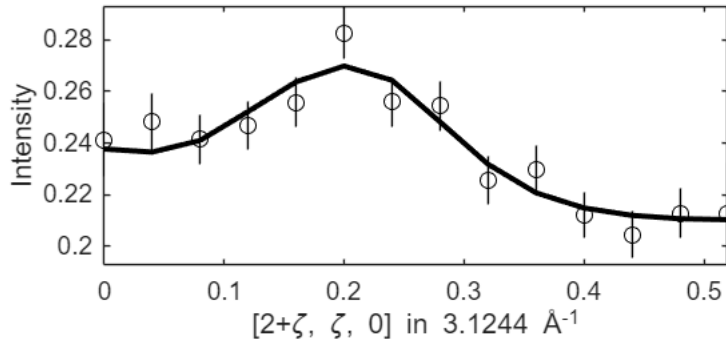
Fe_ei800_align_sym3.sqw
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 88 :
 $\zeta = -0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



 **** En = 103±5 going to 373

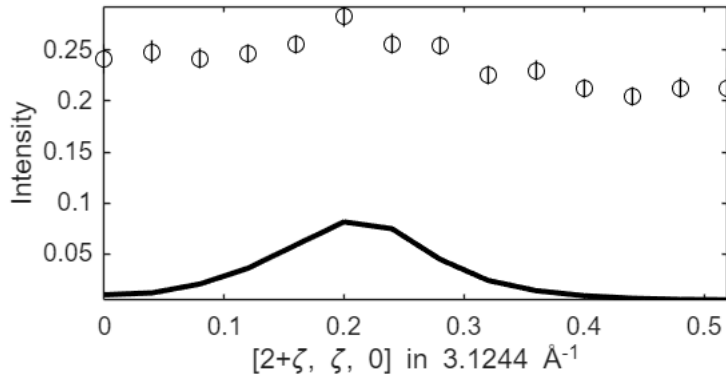
Fe_ei800_align_sym3.sqw

1.2±0.81; J0= 33± 6.3; J1= 0± 0; gamma=69.7325±61.
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 98 $\leq$
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



Fe_ei800_align_sym3.sqw

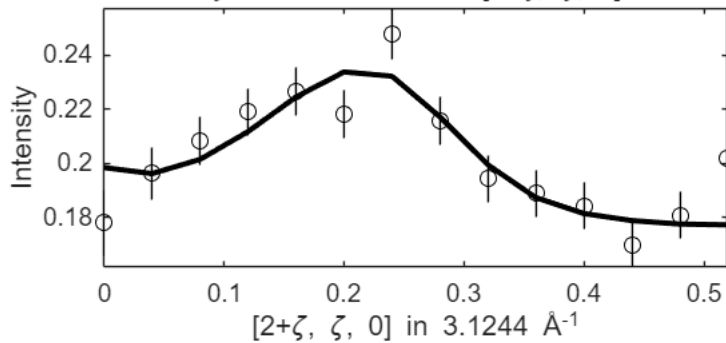
$\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 98 $\leq$
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



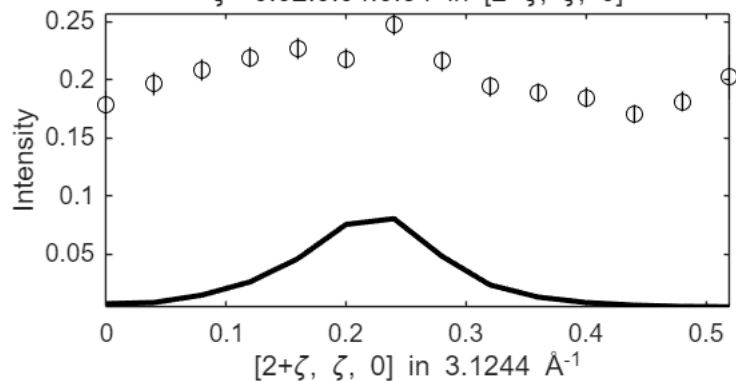
 **** En = 113±5 going to 373

Fe_ei800_align_sym3.sqw

1±0.69; J0= 33± 4.2; J1= 0± 0; gamma=54.0461±50
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 108 $\leq$
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

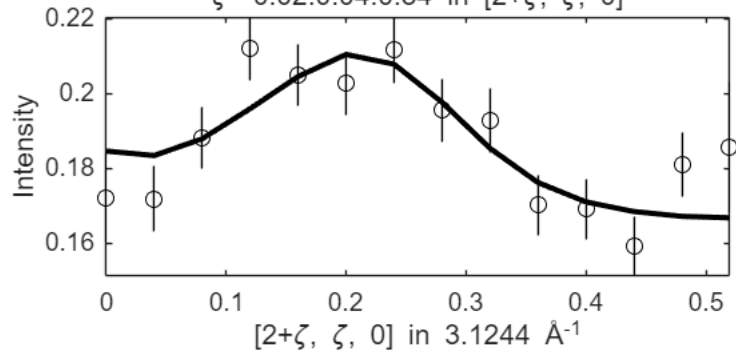


Fe_ei800_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 108
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

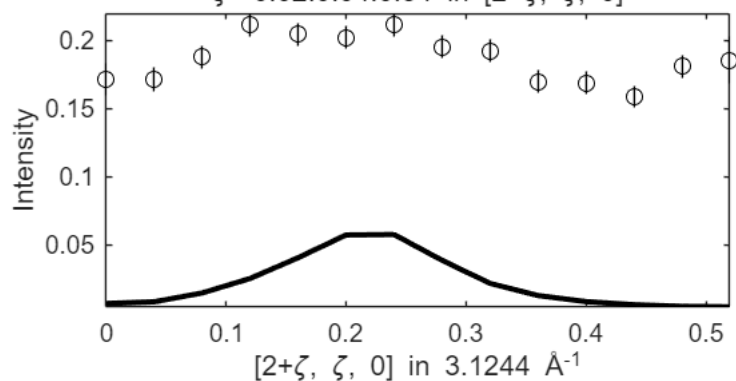


**** En = 123±5 going to 373

Fe_ei800_align_sym3.sqw
 1.1 ± 1.5 ; $J_0 = 38 \pm 14$; $J_1 = 0 \pm 0$; $\gamma = 89.2213 \pm 14$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 118 :
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



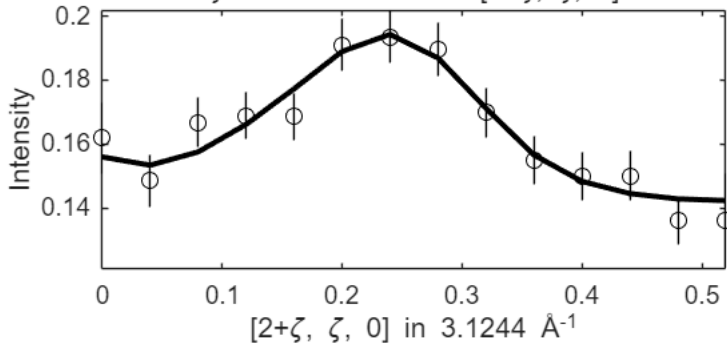
Fe_ei800_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 118 :
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



**** En = 133±5 going to 373

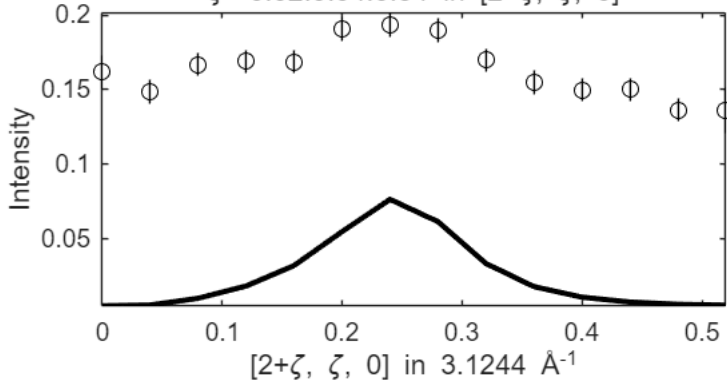
Fe_ei800_align_sym3.sqw

1.1±0.075; J0= 33± 1.2; J1= 0± 0; gamma=57.2961±2.
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 128
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



Fe_ei800_align_sym3.sqw

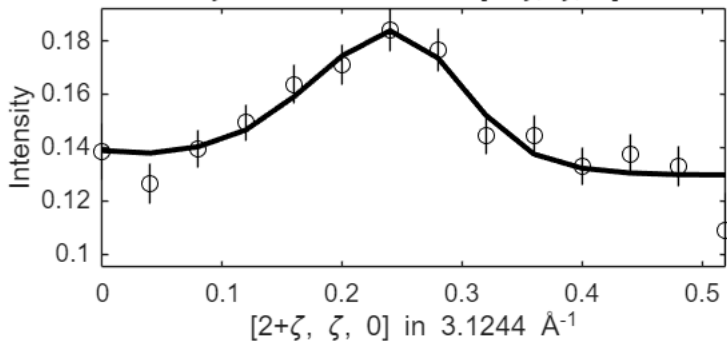
≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 128
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



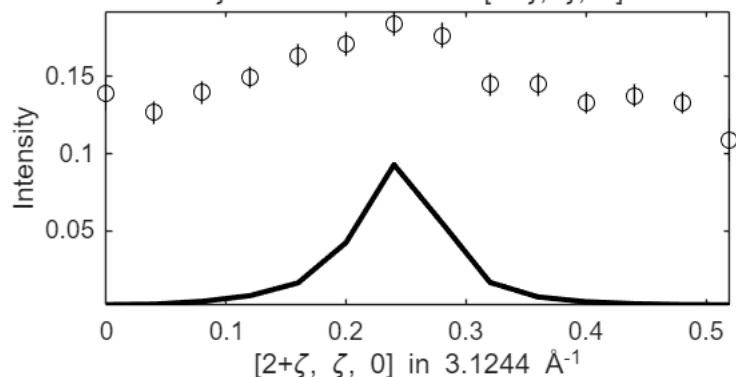
 **** En = 143±5 going to 373

Fe_ei800_align_sym3.sqw

1.69±0.058; J0= 34± 1.3; J1= 0± 0; gamma=26.4576±1.1
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 138
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

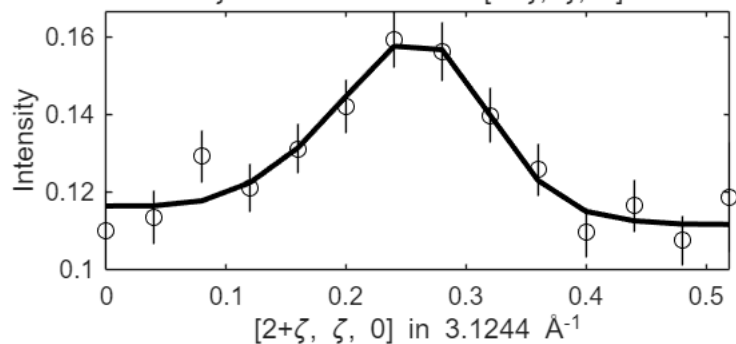


Fe_ei800_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 138
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

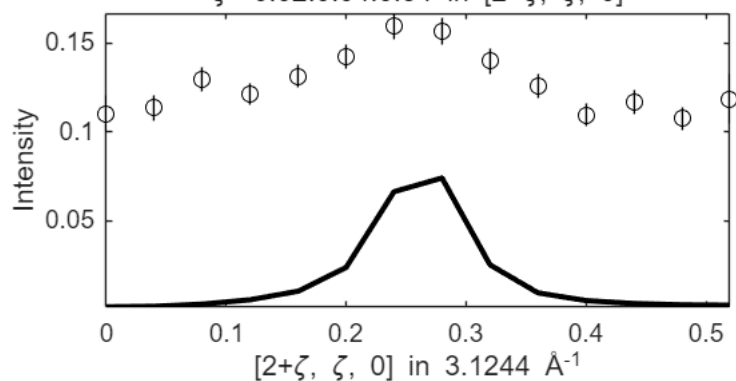


**** En = 153±5 going to 373

Fe_ei800_align_sym3.sqw
 0.63 ± 0.044 ; $J_0 = 33 \pm 0.97$; $J_1 = 0 \pm 0$; $\gamma = 25.8654 \pm 1$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 148
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

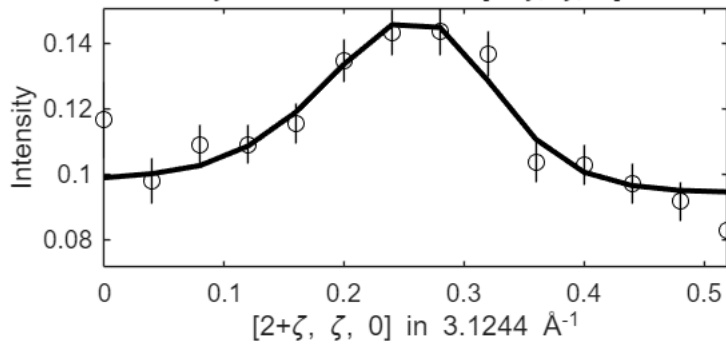


Fe_ei800_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 148
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

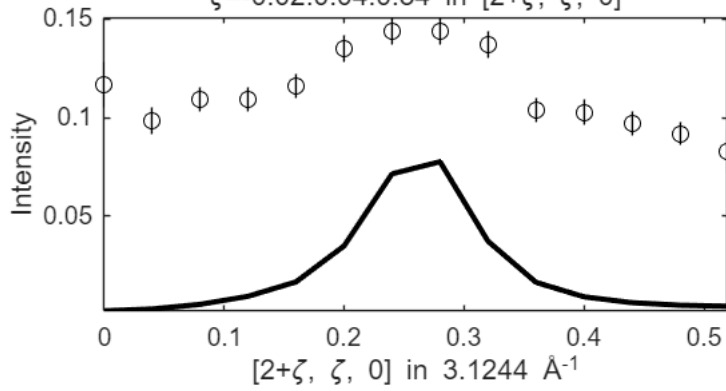


**** En = 163±5 going to 373

Fe_ei800_align_sym3.sqw
 0.9 ± 0.065 ; $J_0 = 35 \pm 1.1$; $J_1 = 0 \pm 0$; $\gamma = 40.0234 \pm 2.0$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 158
 $\zeta = -0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

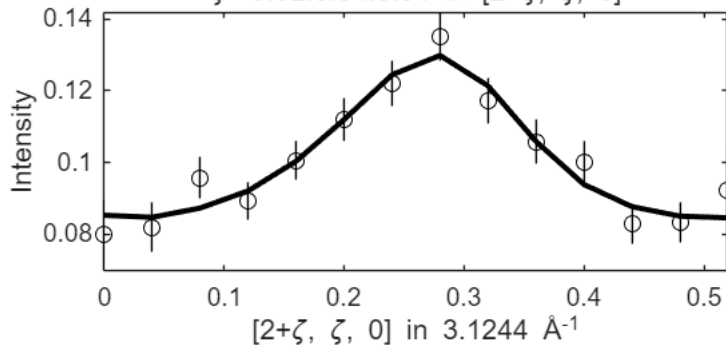


Fe_ei800_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 158
 $\zeta = -0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

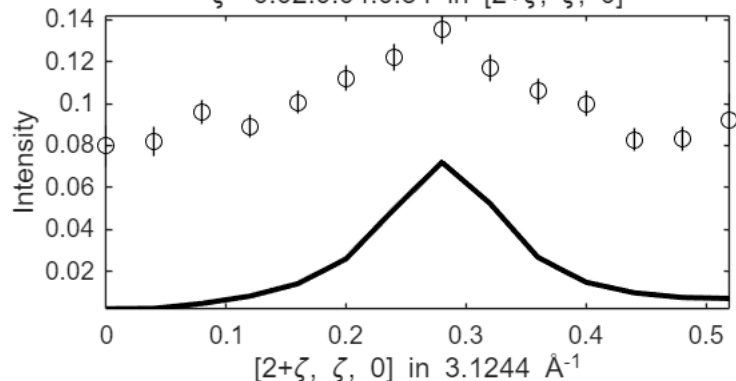


 **** En = 173 ± 5 going to 373

Fe_ei800_align_sym3.sqw
 0.93 ± 0.06 ; $J_0 = 35 \pm 0.89$; $J_1 = 0 \pm 0$; $\gamma = 47.9382 \pm 2.0$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 168
 $\zeta = -0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

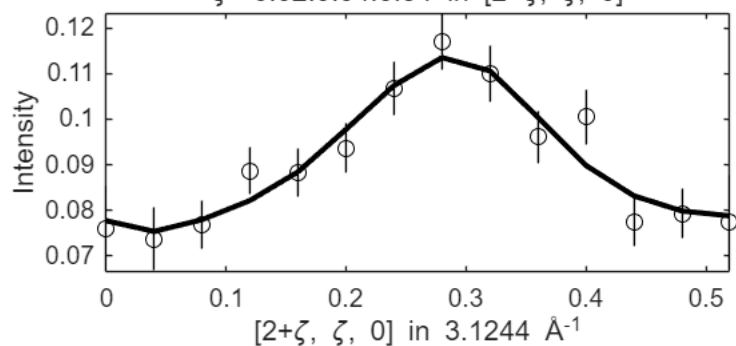


Fe_ei800_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 168
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

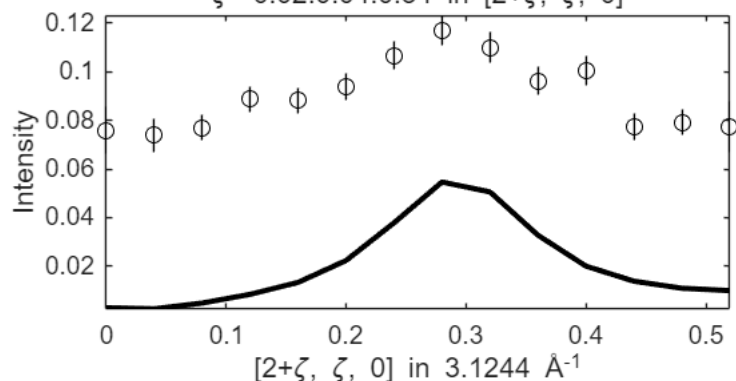


**** En = 183±5 going to 373

Fe_ei800_align_sym3.sqw
 0.94 ± 0.42 ; $J_0 = 35 \pm 1.7$; $J_1 = 0 \pm 0$; $\gamma = 62.2922 \pm 32$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 178
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



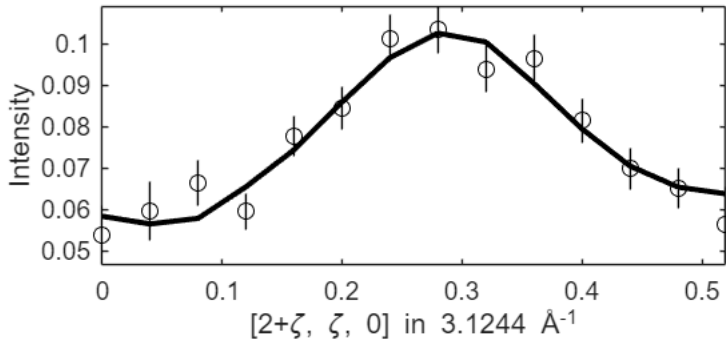
Fe_ei800_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 178
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



**** En = 193±5 going to 373

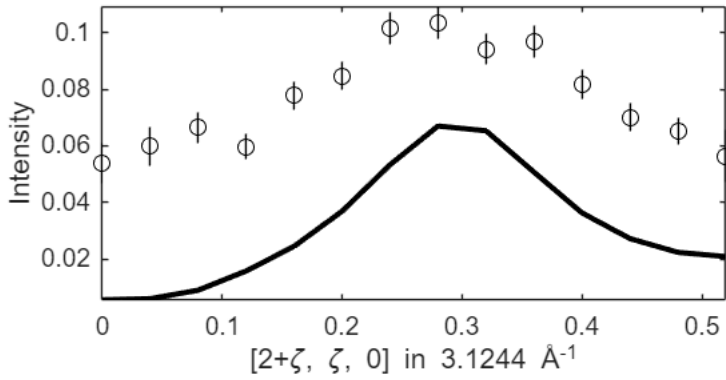
Fe_ei800_align_sym3.sqw

1.8 ± 0.17 ; $J_0 = 38 \pm 0.91$; $J_1 = 0 \pm 0$; $\gamma = 102.079 \pm 3.3$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 188
 $\zeta = -0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



Fe_ei800_align_sym3.sqw

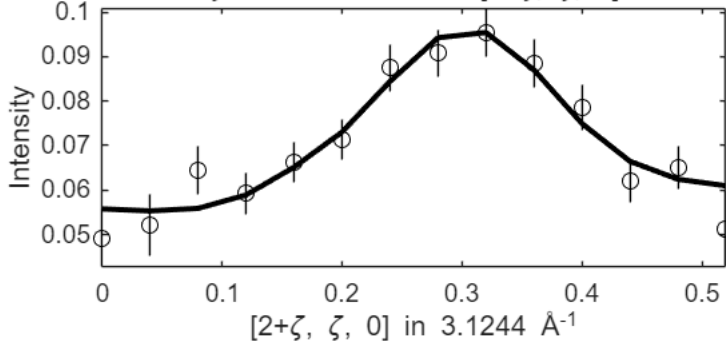
≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 188
 $\zeta = -0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



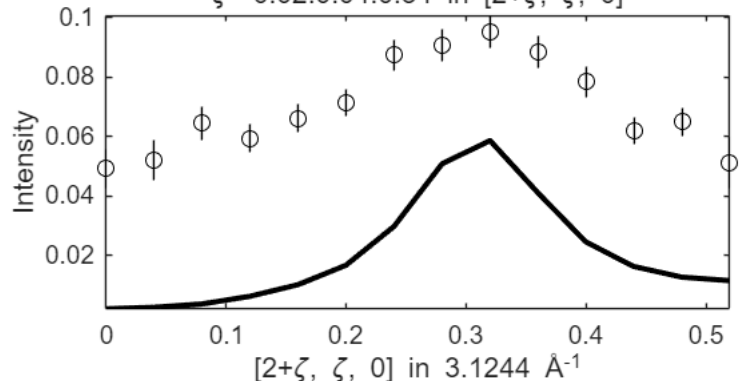
 **** En = 203 ± 5 going to 373

Fe_ei800_align_sym3.sqw

1.93 ± 0.069 ; $J_0 = 35 \pm 0.76$; $J_1 = 0 \pm 0$; $\gamma = 55.6975 \pm 2.2$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 198
 $\zeta = -0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

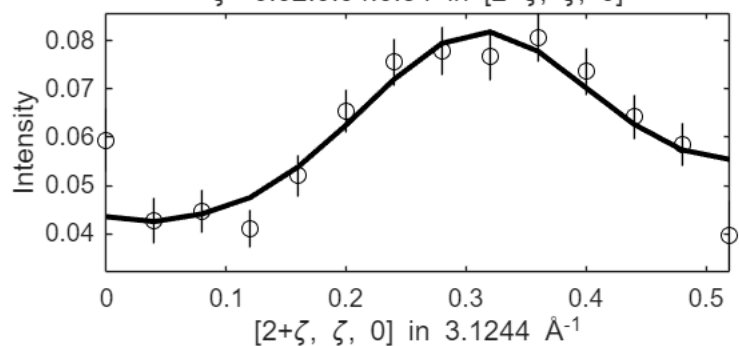


Fe_ei800_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 198 :
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

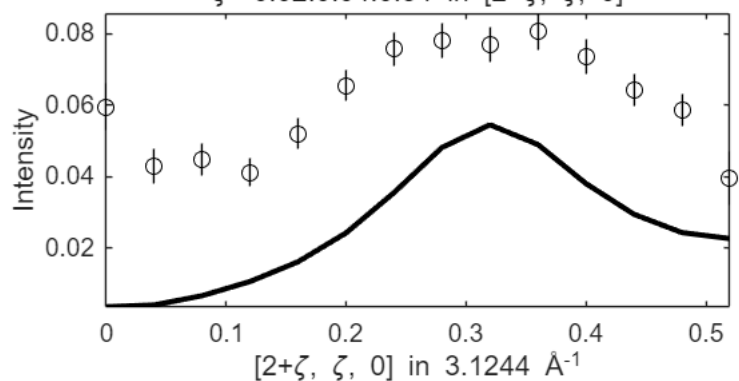


**** En = 213±5 going to 373

Fe_ei800_align_sym3.sqw
 1.5 ± 0.18 ; $J_0 = 37 \pm 1.4$; $J_1 = 0 \pm 0$; $\gamma = 101.563 \pm 5.2$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 208 :
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



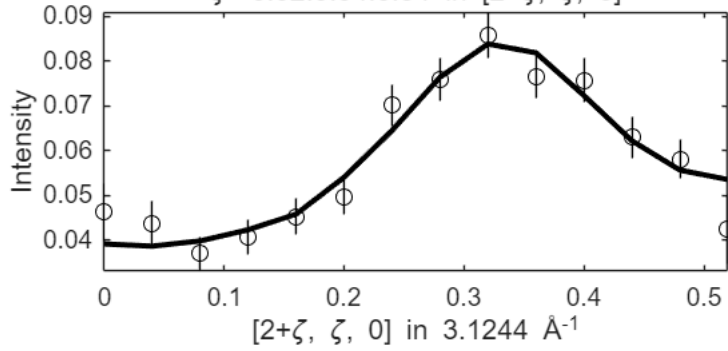
Fe_ei800_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 208 :
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



**** En = 223±5 going to 373

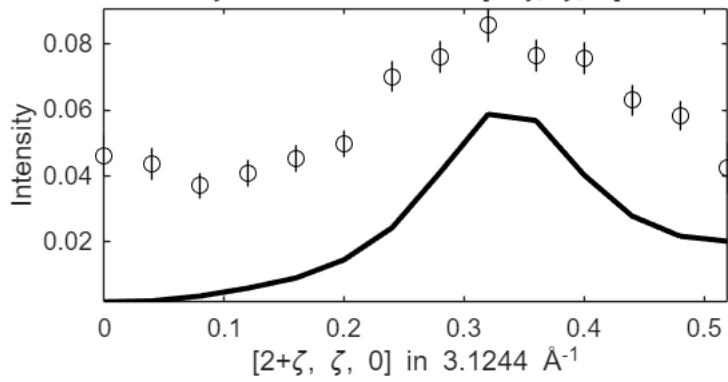
Fe_ei800_align_sym3.sqw

1.1±0.077; J0= 35±0.66; J1= 0± 0; gamma=60.4292±2.
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 218
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



Fe_ei800_align_sym3.sqw

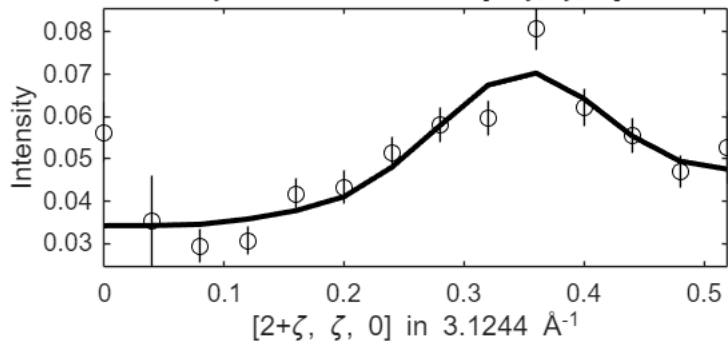
≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 218
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



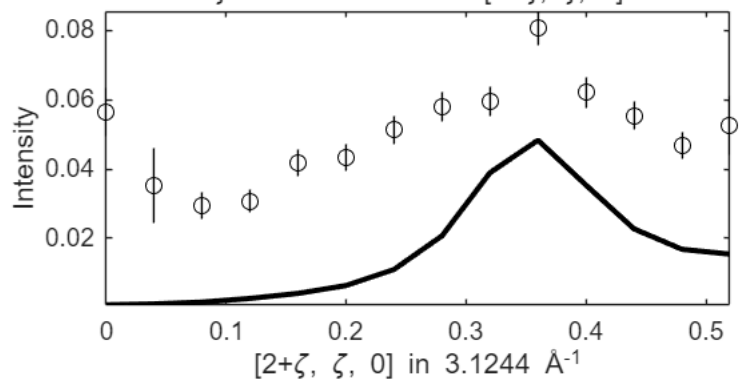
 **** En = 233±5 going to 373

Fe_ei800_align_sym3.sqw

1.67±0.087; J0= 35± 1.1; J1= 0± 0; gamma=43.7314±3.
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 228
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

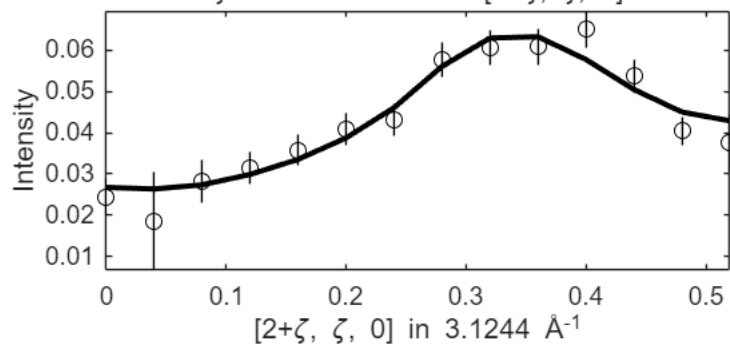


Fe_ei800_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 228 :
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

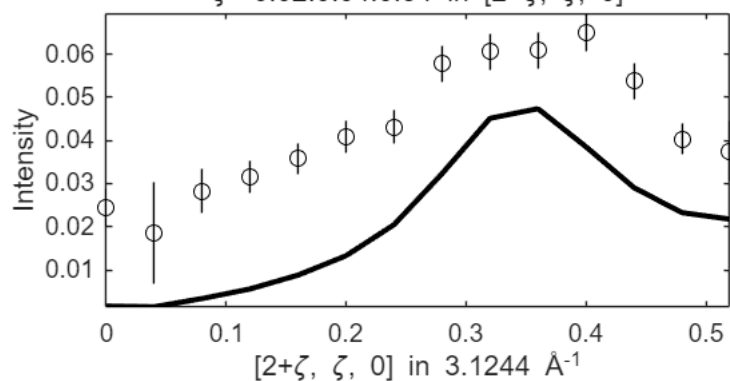


**** En = 243±5 going to 373

Fe_ei800_align_sym3.sqw
 1.1 ± 0.1 ; $J_0 = 37 \pm 0.8$; $J_1 = 0 \pm 0$; $\gamma = 75.4181 \pm 2.9$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 238 :
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



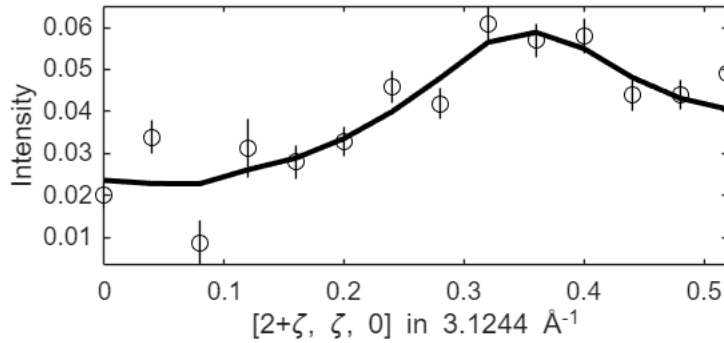
Fe_ei800_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 238 :
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



**** En = 253±5 going to 373

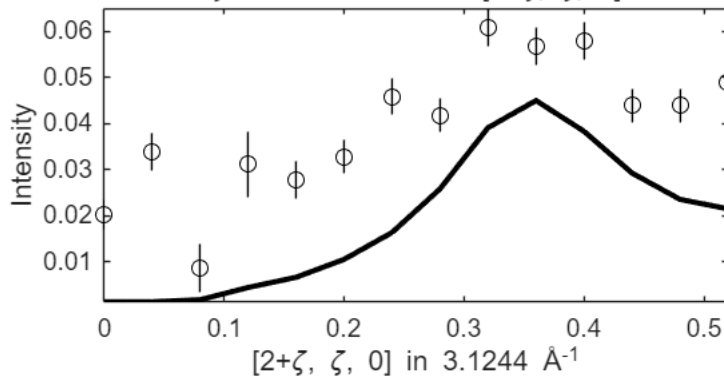
Fe_ei800_align_sym3.sqw

: 0.98 ± 0.16 ; $J_0 = 38 \pm 1.4$; $J_1 = 0 \pm 0$; $\gamma = 70.6268 \pm 4.1$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 248
 $\zeta = -0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



Fe_ei800_align_sym3.sqw

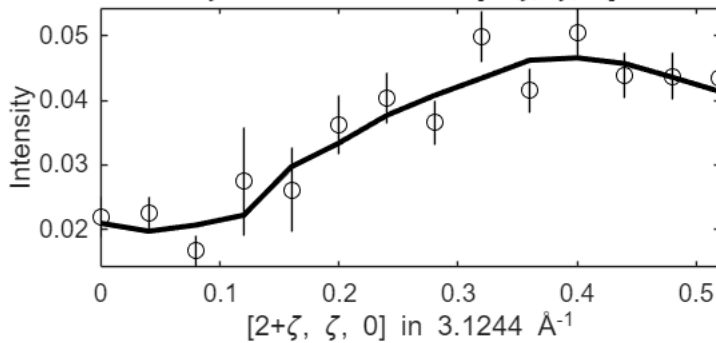
≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 248
 $\zeta = -0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



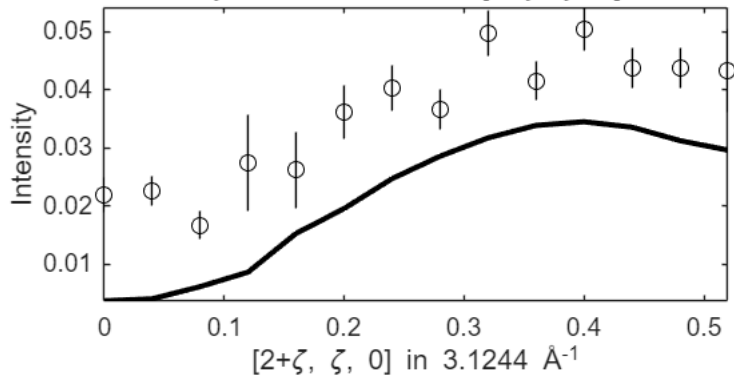
**** En = 263±5 going to 373

Fe_ei800_align_sym3.sqw

: 3.8 ± 2.6 ; $J_0 = 31 \pm 21$; $J_1 = 0 \pm 0$; $\gamma = 290.74 \pm 22.1$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 258
 $\zeta = -0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

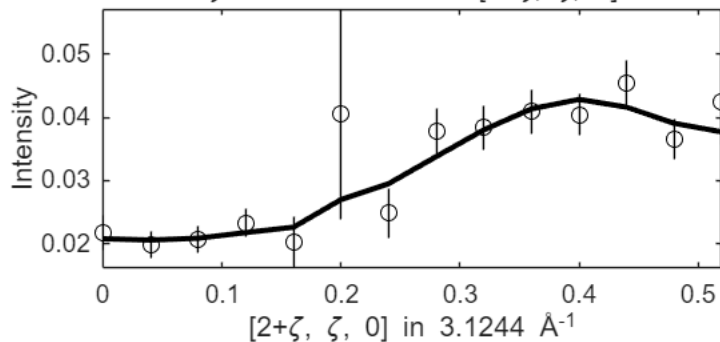


Fe_ei800_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 258
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

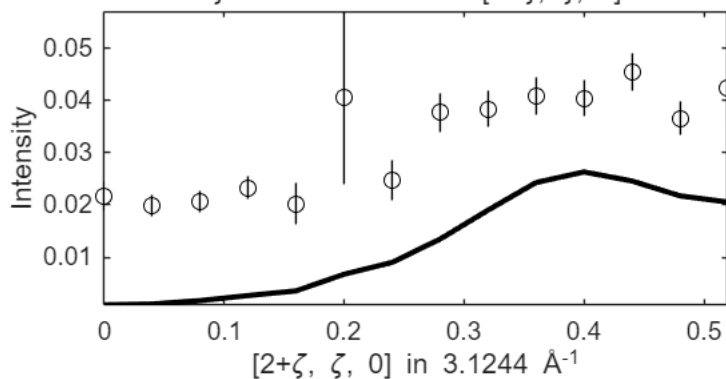


 **** En = 273±5 going to 373

Fe_ei800_align_sym3.sqw
 -0.74 ± 0.31 ; $J_0 = 36 \pm 1.9$; $J_1 = 0 \pm 0$; $\gamma = 82.145 \pm 36$.
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 268
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



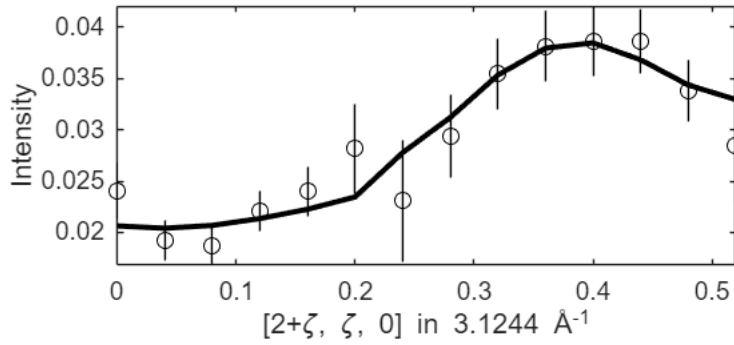
Fe_ei800_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 268
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



 **** En = 283±5 going to 373

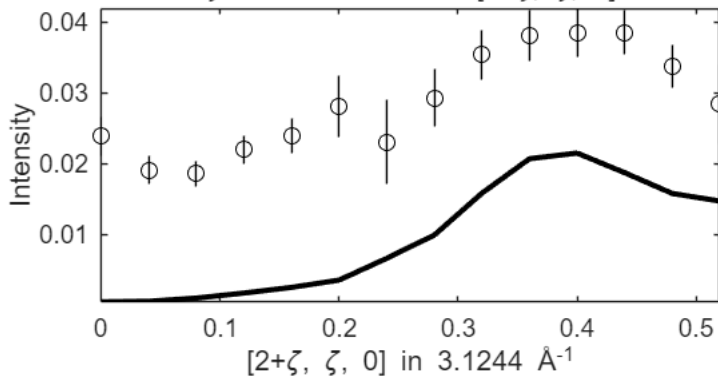
Fe_ei800_align_sym3.sqw

0.52 ± 0.21 ; $J_0 = 39 \pm 1.5$; $J_1 = 0 \pm 0$; $\gamma = 72.2353 \pm 34$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 278
 $\zeta = -0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



Fe_ei800_align_sym3.sqw

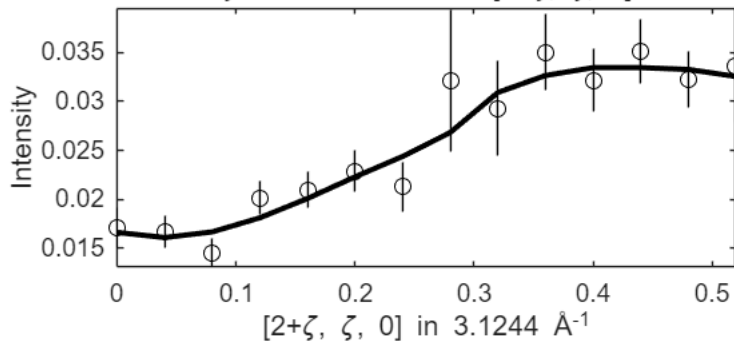
≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 278
 $\zeta = -0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



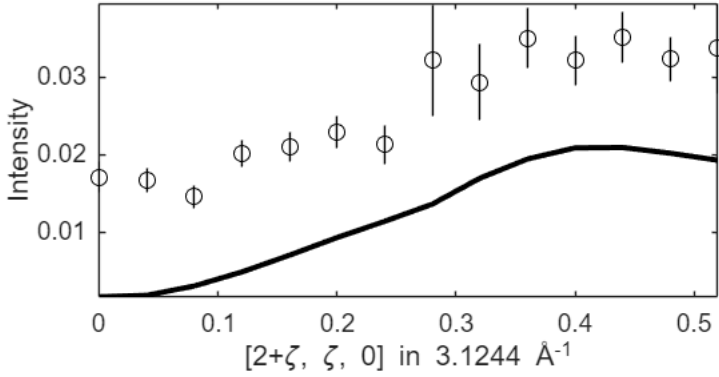
**** En = 293 ± 5 going to 373

Fe_ei800_align_sym3.sqw

$= 2.1 \pm 1.9$; $J_0 = 25 \pm 14$; $J_1 = 0 \pm 0$; $\gamma = 134.957 \pm 3$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 288
 $\zeta = -0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

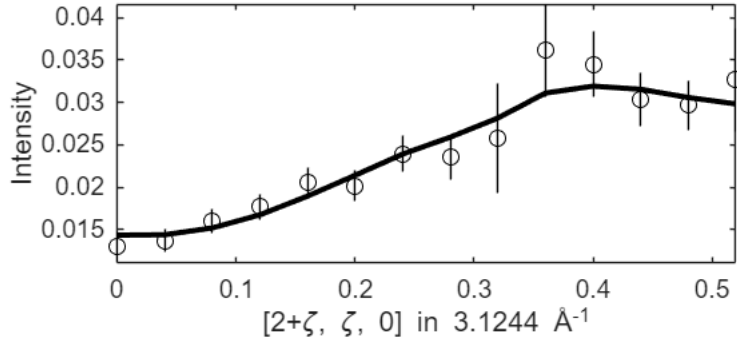


Fe_ei800_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 288
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

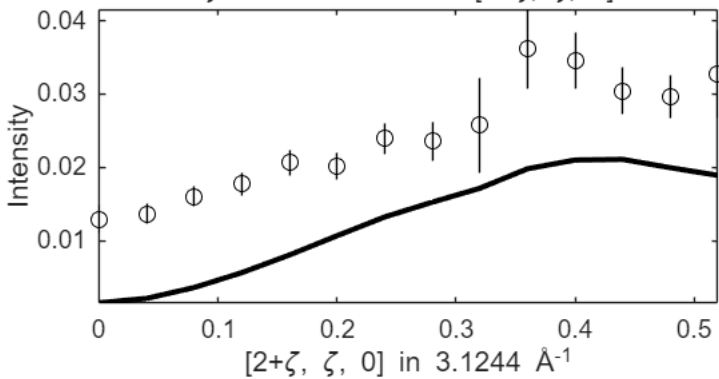


 **** En = 303±5 going to 373

Fe_ei800_align_sym3.sqw
 $= 3 \pm 15$; $J_0 = 21 \pm 85$; $J_1 = 0 \pm 0$; $\gamma = 141 \pm 791$.
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 298
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

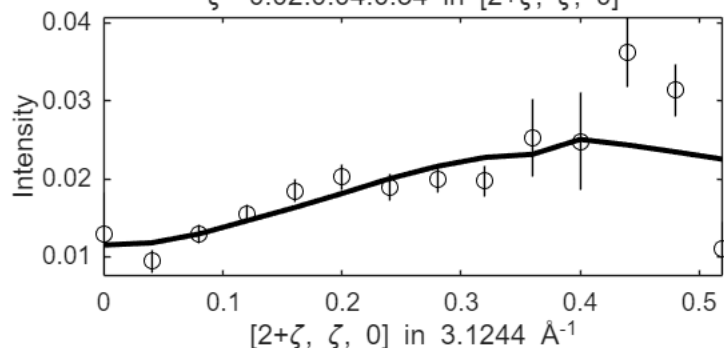


Fe_ei800_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 298
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

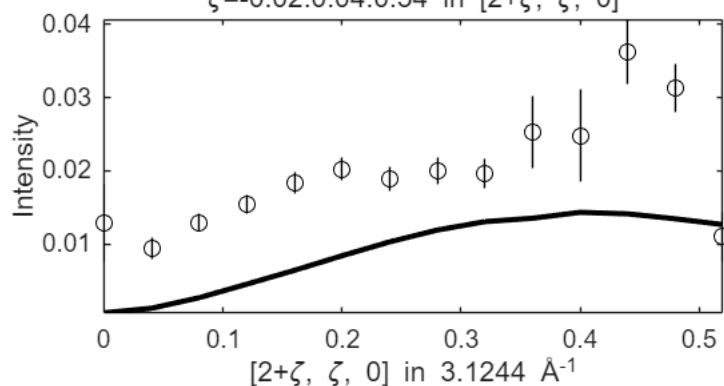


 **** En = 313±5 going to 373

Fe_ei800_align_sym3.sqw
 $1 \pm 2.3 \times 10^2$; $J_0 = 12 \pm 2.1 \times 10^2$; $J_1 = 0 \pm 0$; $\gamma = 90.5343$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 308
 $\zeta = -0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

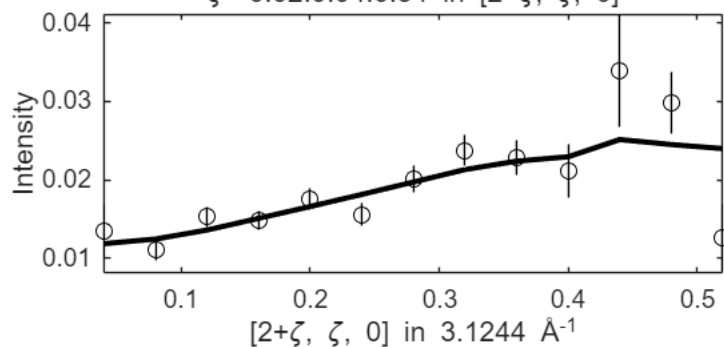


Fe_ei800_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 308
 $\zeta = -0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

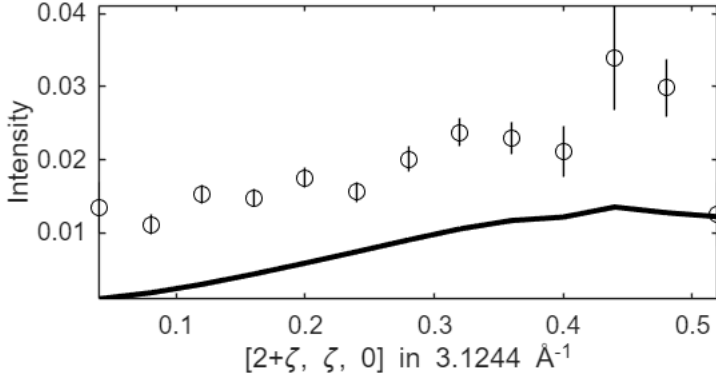


**** En = 323 ± 5 going to 373

Fe_ei800_align_sym3.sqw
 8.4 ± 88 ; $J_0 = 19 \pm 12$; $J_1 = 0 \pm 0$; $\gamma = 20.2323 \pm 15$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 318
 $\zeta = -0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

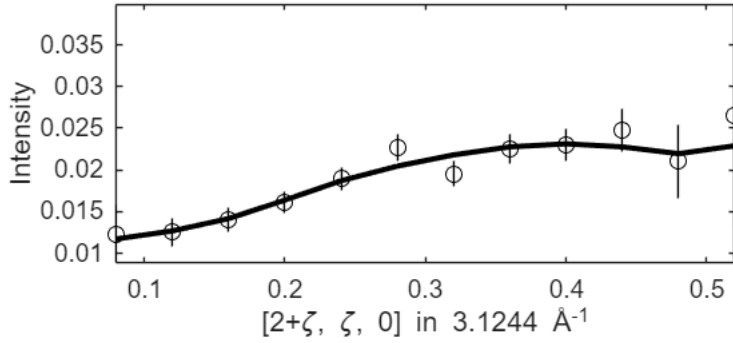


Fe_ei800_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 318
 $\zeta = -0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

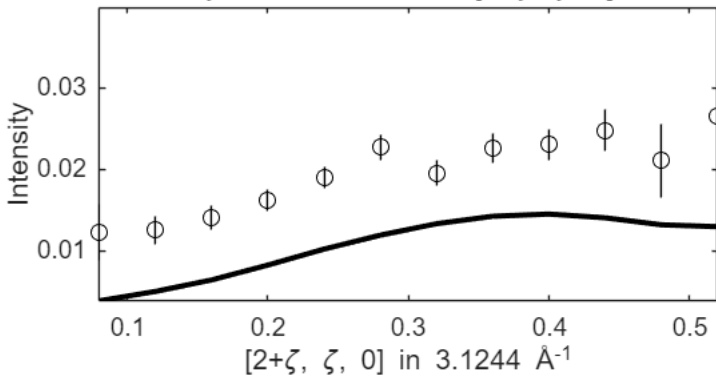


 **** En = 333±5 going to 373

Fe_ei800_align_sym3.sqw
 $16 \pm 2.8e+03$; J0= 12 ± 76 ; J1= 0 ± 0 ; gamma=29.0402±
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 328
 $\zeta = -0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



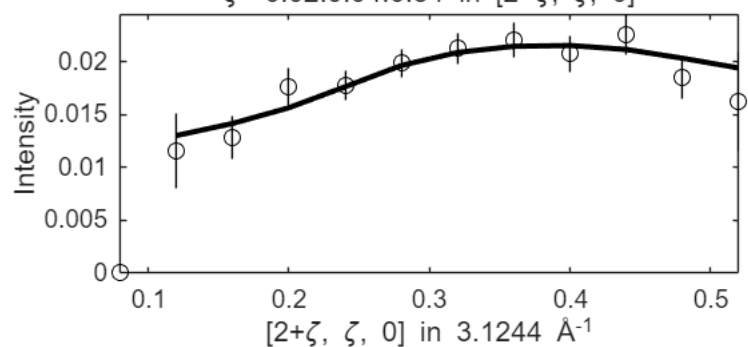
Fe_ei800_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 328
 $\zeta = -0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



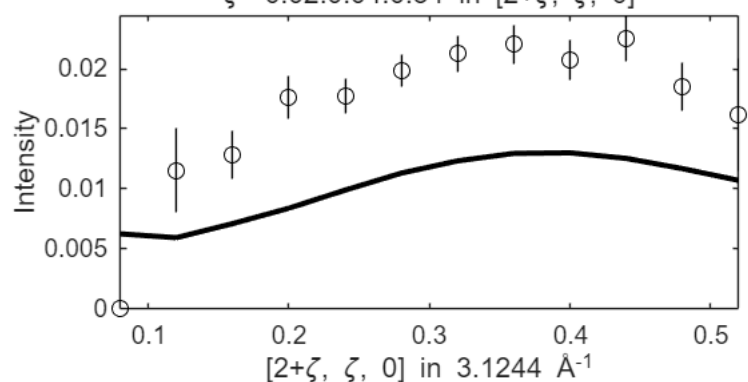
 **** En = 343±5 going to 373

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 7.1429

Fe_ei800_align_sym3.sqw
 $37 \pm 8.6e+03$; $J_0 = 4.3 \pm 83$; $J_1 = 0 \pm 0$; $\gamma = 37.2667 \pm 1$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 338
 $\zeta = -0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

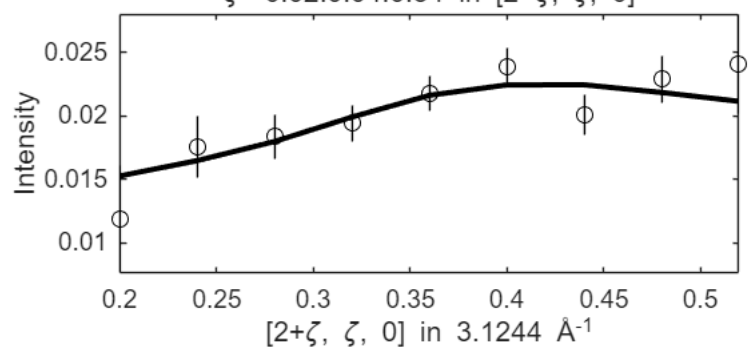


Fe_ei800_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 338
 $\zeta = -0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

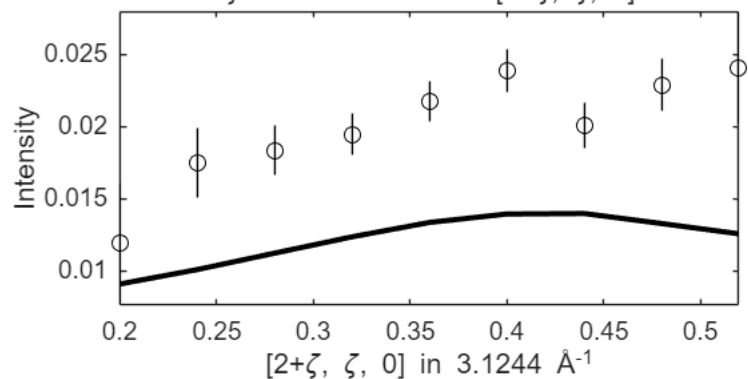


 **** En = 353 ± 5 going to 373

Fe_ei800_align_sym3.sqw
 15 ± 76 ; $J_0 = 15 \pm 10$; $J_1 = 0 \pm 0$; $\gamma = 25.4497 \pm 1$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 348
 $\zeta = -0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

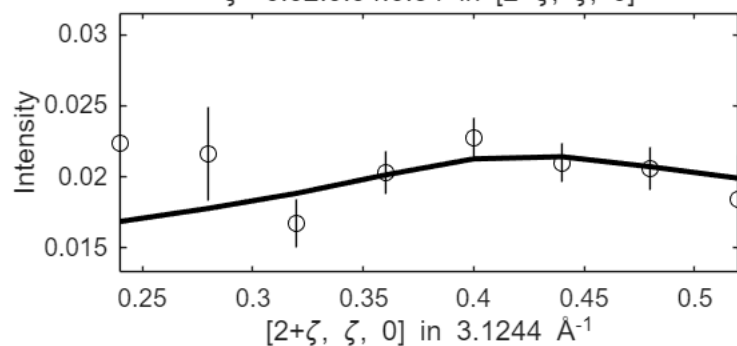


Fe_ei800_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 348
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

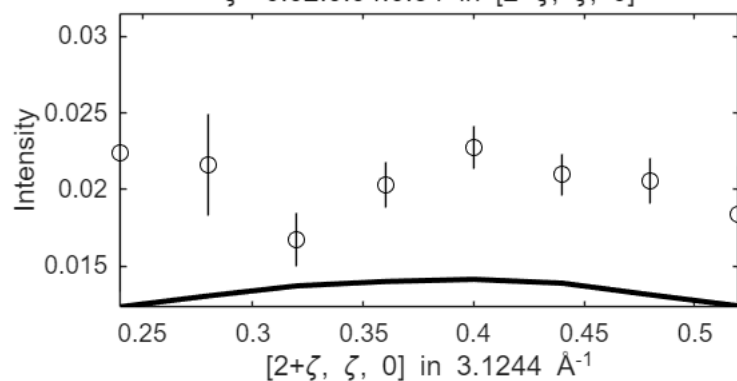


**** En = 363±5 going to 373

Fe_ei800_align_sym3.sqw
 $e+02 \pm 1.2e+05$; $J_0 = 9.7 \pm 19$; $J_1 = 0 \pm 0$; $\gamma = 4.3375$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 358
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

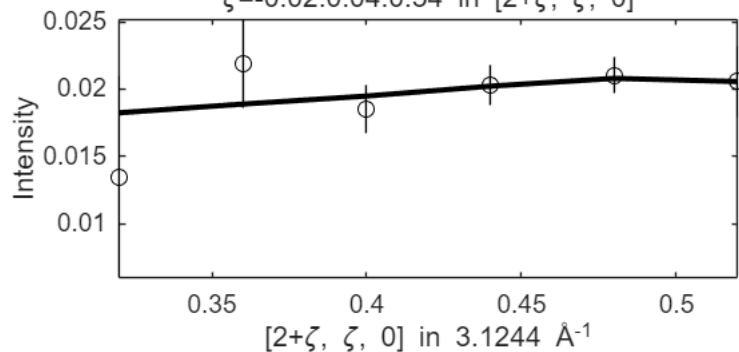


Fe_ei800_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 358
 $\zeta=-0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$

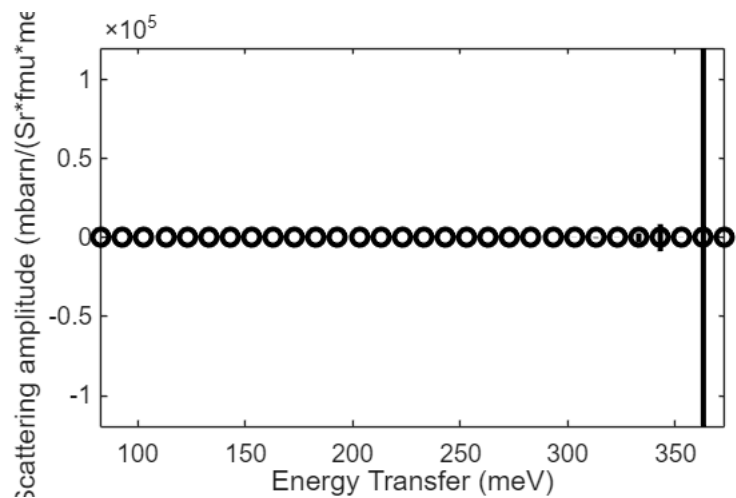
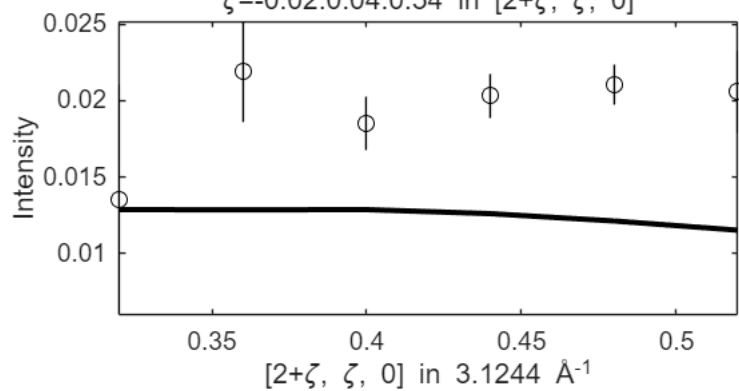


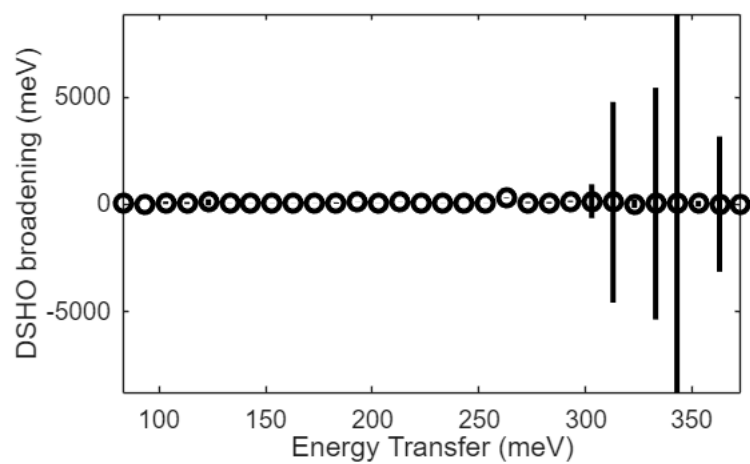
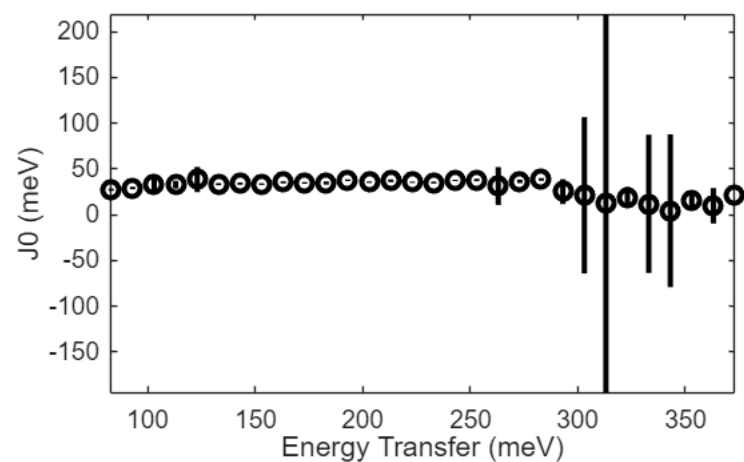
**** En = 373±5 going to 373

Fe_ei800_align_sym3.sqw
 $=2.8e+02 \pm \text{Inf}$; $J_0 = 21 \pm \text{Inf}$; $J_1 = 0 \pm 0$; $\gamma = 0.86587$;
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 368
 $\zeta = -0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$



Fe_ei800_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 368
 $\zeta = -0.02:0.04:0.54$ in $[2+\zeta, \zeta, 0]$





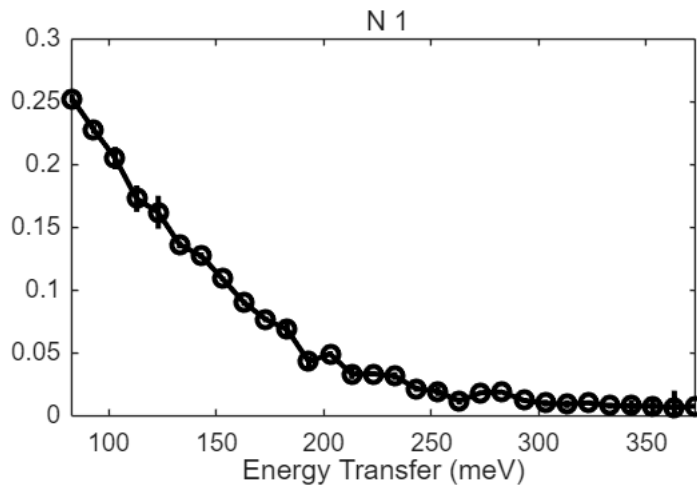
```

c1 = NG_FFCutsFits.(selectCuts);
NG_FFCutsFits.(selectCuts) = [c1{:}];

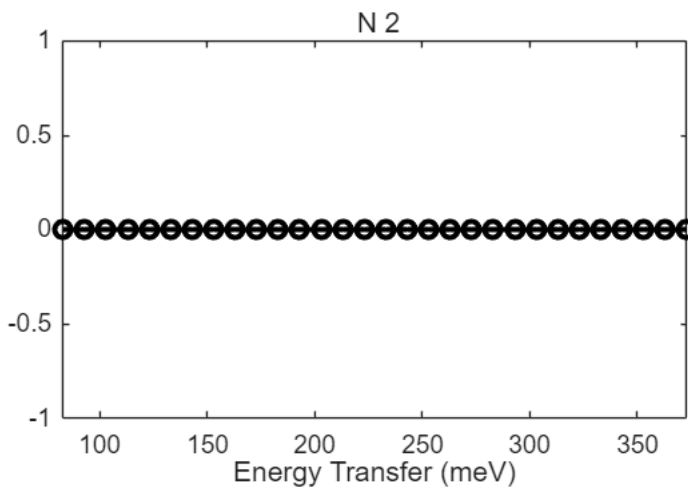
beep
beep
beep

all_fit_par = NG_FFCutsFits.([selectCuts, '_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);ly 0 0.3; keep_figure;

```



```
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure
```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_NG.mlx';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_NG_',selectCuts,'.pdf']);
export(thisFile,targ_file);
%close all;
%end
```

```
fit_src_struct = NG_FFCutsFits;
save(Cutfitfile,"fit_src_struct",-v7.3')
```